

**This assignment will not be graded and is only for practice.**

### Level 1

1) Consider a function  $T : V \rightarrow W$ , where  $V$  and  $W$  are the vector spaces and following statements:

**1 point**

**Statement 1:**  $T(v_1 + v_2) = T(v_1) + T(v_2)$ , where  $v_1, v_2 \in V$ .

**Statement 2:**  $T(cv_1) = cT(v_1)$ , where  $x \in V$  and  $c \in \mathbb{R}$ .

If  $T$  is a linear transformation then  $T$  satisfies

- ☐ statement 1 only.
- ☐ statement 2 only.
- ☐ both the statements 1 and 2.
- ☐ neither statement 1 nor statement 2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

both the statements 1 and 2.

2) Which of the following is/are a linear transformation.

**1 point**

- ☐  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, T(x, y) = (x, x + y, y + 1)$
- ☐  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, T(x, y) = (xy, x + y)$
- ☐  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, T(x, y) = (x + y, y, 2x + y + 1)$
- ☐  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, T(x, y) = (x + y, y, 2x + y)$
- ☐  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, T(x, y) = (\frac{x}{2}, \frac{y}{2})$
- ☐  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, T(x, y) = (\frac{x}{2}, \frac{xy}{2})$

No, the answer is incorrect.

Score: 0

## Accepted Answers:

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, T(x, y) = (x + y, y, 2x + y)$$

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, T(x, y) = \left(\frac{x}{2}, \frac{y}{2}\right)$$

3) Let  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$  be a linear transformation. Then which of the following options is /are true? **1 point**

☐  $T(v_1 + v_2) = T(v_1) + T(v_2)$ , where  $v_1, v_2 \in \mathbb{R}^3$ .

☐  $T(0, 0, 0) = (0, 0, 0, 0)$

☐  $T(0, 0, 0) = (1, 1, 1, 1)$

☐  $T(3(1, 2, 0)) = 3T(1, 2, 0)$

No, the answer is incorrect.

Score: 0

## Accepted Answers:

$$T(v_1 + v_2) = T(v_1) + T(v_2), \text{ where } v_1, v_2 \in \mathbb{R}^3.$$

$$T(0, 0, 0) = (0, 0, 0, 0)$$

$$T(3(1, 2, 0)) = 3T(1, 2, 0)$$

4) Let  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$  be a linear transformation, defined as  $T(x, y, z) = (x + y, y + z)$ . Then which of the following options is /are true? **1 point**

☐  $T(0, 0, 0) = (0, 0)$

☐  $T(1, 2, 3) = (3, 5)$

☐  $T(0, 1, 1) = (1, 2)$

☐  $T(1, 1, 1) = (3, 2)$

☐  $T$  is injective.

No, the answer is incorrect.

Score: 0

## Accepted Answers:

$$T(0, 0, 0) = (0, 0)$$

$$T(1, 2, 3) = (3, 5)$$

$$T(0, 1, 1) = (1, 2)$$

5) Choose the correct set of options:

**1 point**

- ☐  $T$  is a one to one linear transformation if and only if there does not exist any  $v = 0 \in V$  so that  $T(v) = 0$ .
- ☐ If  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a surjective linear transformation, then it cannot be injective.
- ☐ If  $T : \mathbb{R} \rightarrow \mathbb{R}$  is a linear transformation which is not injective, then  $T$  must be 0, i.e.,  $T(v) = 0$  for all  $v \in \mathbb{R}$ .
- ☐ If there exists some non-zero vector  $v \in V$ , such that  $T(v) = 0$ , for a linear transformation  $T : V \rightarrow W$ , then  $T$  cannot be an isomorphism.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$  is a one to one linear transformation if and only if there does not exist any  $v = 0 \in V$  so that  $T(v) = 0$ .

If  $T : \mathbb{R} \rightarrow \mathbb{R}$  is a linear transformation which is not injective, then  $T$  must be 0, i.e.,  $T(v) = 0$  for all  $v \in \mathbb{R}$ .

If there exists some non-zero vector  $v \in V$ , such that  $T(v) = 0$ , for a linear transformation  $T : V \rightarrow W$ , then  $T$  cannot be an isomorphism.

6) If  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a linear transformation, such that  $T(x, y) = (x, 0)$ , then which of the following options is true?

**1 point**

- ☐  $T$  is both one to one and onto.
- ☐  $T$  is one to one, but not onto.
- ☐  $T$  is onto, but not one to one.
- ☐  $T$  is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$  is neither one to one nor onto.

7) If  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a linear transformation, such that  $T(x, y) = (x + y, x - y)$ , then which of the following options is true? **1 point**

- ☐  $T$  is both one to one and onto.
- ☐  $T$  is one to one, but not onto.
- ☐  $T$  is onto, but not one to one.
- ☐  $T$  is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$  is both one to one and onto.

8) Let  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be a linear transformation, such that  $T(2, 1) = (3, 2)$  and  $T(1, 3) = (0, 0)$ . **1 point**  
Which of the following options is true?

- ☐  $T(x, y) = \frac{1}{5}(3x, 2y)$
- ☐  $T(x, y) = (x + 1, y + 1)$
- ☐  $T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$
- ☐  $T$  cannot be determined from the given information.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$

9) Consider the linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined as  $T(v) = Av$ , where  $v = \begin{bmatrix} x \\ y \end{bmatrix}$ , and  $A = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$ . Which of the following options is true? **1 point**

- ☐  $T$  is an isomorphism.
- ☐  $T$  is one to one, but not onto.
- ☐  $T$  is onto, but not one to one.
- ☐  $T$  is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$  is an isomorphism.

## Level 2

10) Let  $S : V_1 \rightarrow V_2$  and  $T : V_2 \rightarrow V_3$  be two linear transformations. Let us define  $T \circ S : V_1 \rightarrow V_3$  by  $T \circ S(v) = T(S(v))$ . Choose the correct set of options. **1 point**

- ☐ If  $T \circ S$  is injective, then  $T$  must be injective.
- ☐ If  $T \circ S$  is injective, then  $S$  must be injective.
- ☐ If  $T \circ S$  is surjective, then  $T$  must be surjective.
- ☐ If  $T \circ S$  is surjective, then  $S$  must be surjective.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If  $T \circ S$  is injective, then  $S$  must be injective.  
If  $T \circ S$  is surjective, then  $T$  must be surjective.

11) If  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$  is a linear transformation defined by  $T(x, y) = (2x + 3y, 5x - y, x + 6y)$ . **1 point**  
Which of the following options is true?

- ☐  $T$  is both one to one and onto.

- ☐  $T$  is one to one, but not onto.
- ☐  $T$  is onto, but not one to one.
- ☐  $T$  is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$  is one to one, but not onto.

12) If  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  is a linear transformation defined as follows,  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  **1 point**

$T(x, y, z) = (a_1x + b_1y + c_1z, a_2x + b_2y + c_2z, a_3x + b_3y + c_3z)$ . This can be represented as  $T(v) = Av$ , where  $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ , and  $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ . Which of the following options are true?

- ☐  $T$  is injective if and only if the system of linear equations  $Av = 0$  has a unique solution.
- ☐  $T$  is injective if and only if  $\text{rank}(A) = 3$ .
- ☐ If the system of linear equations  $Av = b$  for any  $b \in \mathbb{R}^3$ , has a solution, then  $T$  must be surjective.
- ☐ If  $T$  is not surjective, then there exists some  $b \in \mathbb{R}^3$ , such that the system of linear equations  $Av = b$  has no solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$  is injective if and only if the system of linear equations  $Av = 0$  has a unique solution.

$T$  is injective if and only if  $\text{rank}(A) = 3$ .

If the system of linear equations  $Av = b$  for any  $b \in \mathbb{R}^3$ , has a solution, then  $T$  must be surjective.

If  $T$  is not surjective, then there exists some  $b \in \mathbb{R}^3$ , such that the system of linear equations  $Av = b$  has no solution.

**Your score is: 0/12**